Revenue Guard: Automated Anomaly Detection Agent

Ensuring Financial Integrity through High-Performance Data Auditing.

I built this agent to act as a first line of defense against financial leaks and transaction errors

The Problem: High-volume transaction data (50,000+ rows) often hides "silent" revenue leaks.

Constraints: Large datasets can crash standard office laptops (like an i5 with 16GB RAM) if not optimized.

The Need: A tool that is fast, secure against SQL injection, and memory-efficient.

Memory Optimization: High-Efficiency Data Handling

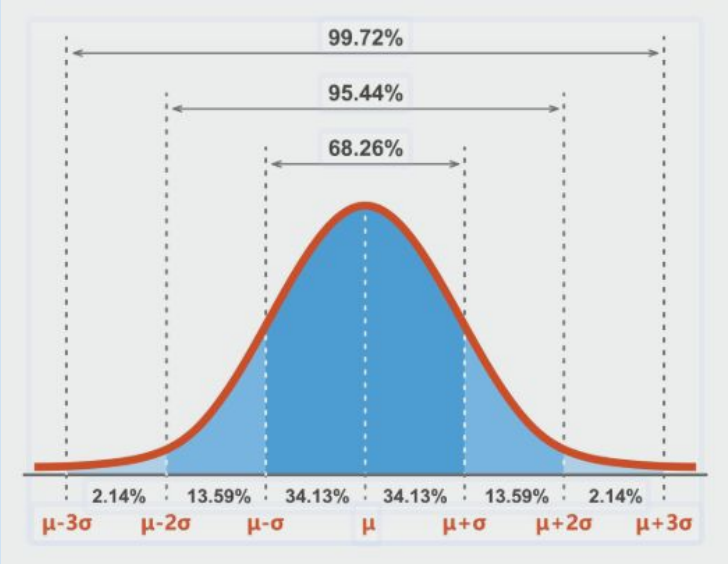
- Large-scale transaction datasets (50k+ rows) often exceed the efficiency limits of standard RAM, causing system lag.
- Implemented **Data Downcasting** (e.g., converting `int64` to `int32` and `float64` to `float32`).
- Achieved a **50% reduction in memory footprint**, ensuring seamless performance on mid-range hardware

Database Integration: Enterprise-Grade Security

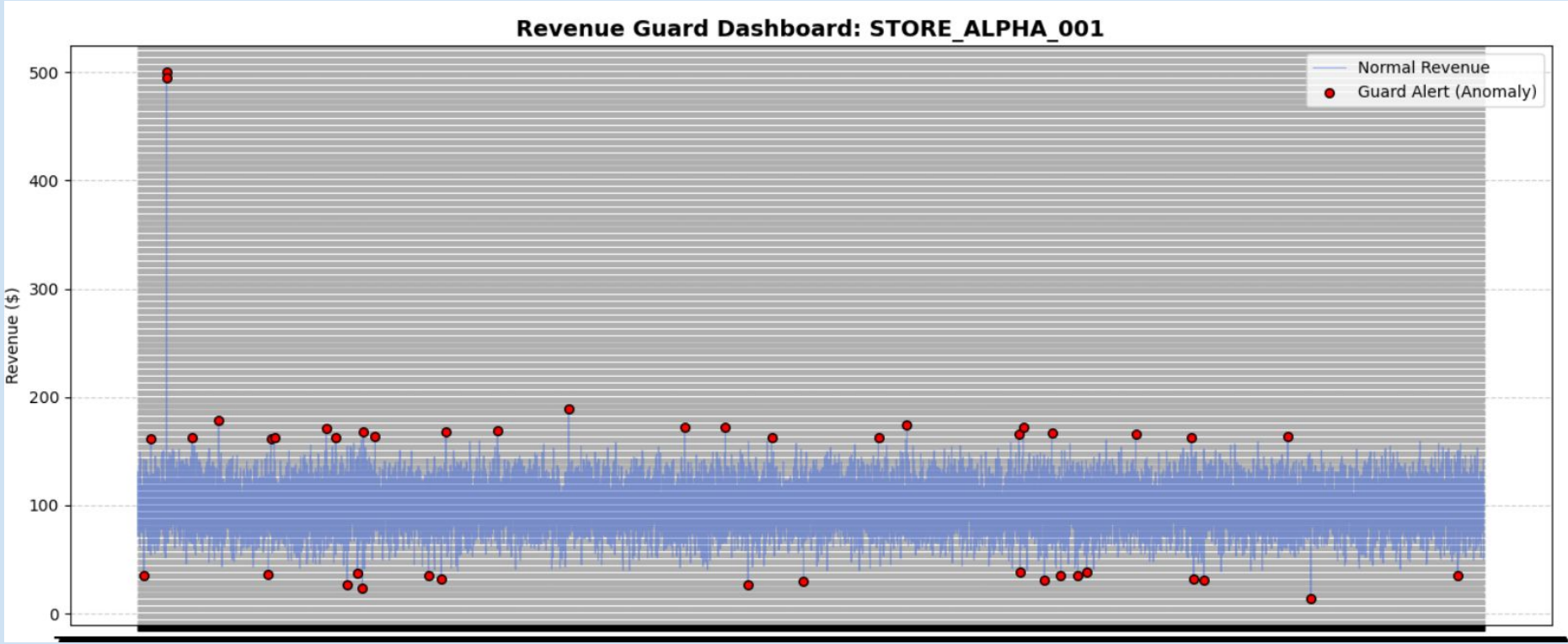
- Migrated volatile data to a persistent **SQLite** backend to decouple storage from the active analysis environment.
- Employed **Parameterized SQL Queries** (using the `?` placeholder) as a critical safeguard against SQL injection attacks, adhering to secure coding standards.

Statistical Anomaly Detection

- **Logic:** Utilized **Z-Score (3 sigma)** statistical modeling to calculate deviations from the moving mean.
- **Isolation:** Anomalies are defined as data points falling beyond 3.0 standard deviations—effectively separating "noise" from true financial risks.



The agent successfully scanned 50,000 transactions and flagged **43 high-risk anomalies** in sub-second time



Summary:

- Delivered a functional end-to-end pipeline covering data generation, secure storage, and automated auditing.
- Proven ability to handle enterprise-level datasets (50,000+ records) on standard professional hardware through meticulous memory management.
- Provided clear, data-driven visualizations that transform complex transaction logs into actionable business intelligence.

Github: https://github.com/Junaid1991-maker/Revenue-Guard-Agent/tree/main

Linkedin: https://www.linkedin.com/in/m-junaid-iqbal-5644b882/